

# RAHUL MITRA

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## EDUCATION

### National University of Singapore

Singapore

*Bachelor of Engineering, Industrial Systems Engineering*

Aug 2023 – Jul 2027

- Pursuing a Second Major in Computer Science and a Minor in Mathematics.
- Awarded Certificate of Outstanding Performance in 3D Computer Vision (CS4277) — top student in a class of 24, NUS School of Computing.
- Relevant coursework: Artificial Intelligence, 3D Computer Vision, Software Engineering, Simulation, Statistics, Operations Research, Linear Algebra, Probability.

## EXPERIENCE

### STMicroelectronics

Singapore

*Operations Research (NUS System Design Project), Contract*

Aug 2026 – Present

- Build an AI-driven put-away recommendation system for STMicroelectronics' Singapore warehouse, combining picking history, demand forecasts, and live capacity to replace experience-based slotting decisions.

### Amazon

Singapore

*Robotics Vision Engineer (BlendED AI+X)*

Jul 2026 – Present

- Build vision systems for camera ISP enhancement, super-resolution, and image restoration under motion and low light.
- Engineer processing and evaluation stages that deliver perception-ready frame quality for downstream robotics vision.

### Abbott Laboratories

Singapore

*Operational AI Systems & Data Engineer (Contract)*

Jun 2026 – Present

- Build AI-enabled systems and data workflows for process improvement, manufacturing analytics, operational decision support, and supply-chain automation.

### Abbott Laboratories

Singapore

*Data Analytics and Project/Product Management Intern*

Jan 2026 – Jun 2026

- Developed a SimPy discrete-event digital twin and OR-Tools CP-SAT optimizer for hybrid flow-shop scheduling, translating production objectives and constraints into decision-ready schedules.
- Delivered practical AI upskilling to the regional engineering workforce for day-to-day operational use.

### People's Association

Singapore

*Software Engineer (Citizen Developer)*

Sep 2025 – Present

- Built an end-to-end digital garden-plot balloting platform spanning admin and public frontends, backend, GIS mapping, and deployment, now production-ready for national rollout across Singapore.

## PROJECTS

### Volt Pulse SG — SMU Hack For Cities 2026 (Top 8 of 50+ teams)

Jan 2026

- Built an agentic AI system for tracking HDB household energy costs using SEALION embeddings, Supabase vector search, RRF routing, and scheduler-agent planning.

### Maritime Deficiency Severity Forecasting — Maritime Hackathon 2025 (Team Lead)

2025

- Led model training for a maritime-inspection NLP system, fine-tuning a 109M-parameter BERT with DNN layers, gradient clipping, and hyperparameter optimization on the ASPIRE 2A supercomputer.

### AgeWellLah.AI — NUS HealthHack 2025

2025

- Built an elderly-care triage chatbot using OpenAI GPT-4, InterSystems IRIS Vector Search, and Flask, with OAuth2-secured patient profiles and PDPA-aware design.

## LEADERSHIP AND ACTIVITIES

### NTUC Health Active Ageing Centre

Singapore

*Elderly Care Volunteer*

Sep 2025 – Present

- Support weekly senior-engagement activities, deepening empathy for healthcare, accessibility, and human-centered community product design.

## SKILLS AND CERTIFICATIONS

**Technical:** Python, BERT, DNN, RAG, AI agents, GPT-4, Gemini, SEALION, vector search, model evaluation, NLP; vision: camera ISP, super-resolution, image restoration, SfM, epipolar geometry, multi-view stereo; HPC: ASPIRE 2A, CUDA

**Certifications:** NVIDIA Disaster Risk Monitoring with Satellite Imagery; Deep Reinforcement Learning Hands-On (Packt)